LIFTING A FORMALLY SMOOTH COMPLETE LOCAL ALGEBRA ALONG A RESIDUE FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Existence of the diagram of Noetherian complete local rings with a power series ring over a Cohen ring at the top. Lemma \ref{lemma-embed-map-Noetherian-complete-local-rings}, cited at proof lines 10266 and 10295. Provides the ambient power series ring S over R whose special fibre is the given algebra.
- The kernel of a surjection of regular local rings is generated by part of a regular system of parameters. Algebra, Lemma \ref{algebra-lemma-regular-quotient-regular}, cited at proof lines 10277-10279. Identifies generators x-bar_1, ..., x-bar_r of the kernel that extend to a regular system of parameters.
- The quotient of a power series ring by part of a regular system of parameters is a power series ring over a field or a Cohen ring. Lemma \ref{lemma-quotient-power-series-ring-over-Cohen}, cited at proof lines 10288-10290. Identifies the cut ring as a power series ring, so that the flat-fibre criterion applies to it.
- Formal smoothness of a local map whose special fibre is formally smooth, and base change of formal smoothness. Proposition \ref{proposition-fs-flat-fibre-fs} and Lemma \ref{lemma-base-change-fs}, cited at proof lines 10300-10304. Give formal smoothness of R to S once the special fibres agree, and of the base change C = S tensor A over A.

Source reference. Original lemma 07NR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9771–9771. The lemma's own canonical tag, quoted

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so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9768–9768. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00NR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 869–869. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9767–9767. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07NQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9770–9770. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07EG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9446–9446. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let A be a Noetherian complete local ring with residue field k. Let B be a Noetherian complete local k-algebra. Assume $k \to B$ is formally smooth in the $\mathfrak{m}_B$-adic topology. Then there exists a Noetherian complete local ring C and a local homomorphism $A \to C$ which is formally smooth in the $\mathfrak{m}_C$-adic topology such that $C \otimes_A k \cong B$.*

Proof. Choose a diagram

$$\begin{array}{ccc} S & \longrightarrow & B \\ \uparrow & & \uparrow \\ R & \longrightarrow & A \end{array}$$

as in Lemma 07NN. Let $t_1, \ldots, t_d$ be a regular system of parameters for R with $t_1 = p$ in case the characteristic of k is $p > 0$. As B and $\overline{S} = S \otimes_R k$ are regular we see that $\mathrm{Ker}(\overline{S} \to B)$ is generated by elements $\overline{x}_1, \ldots, \overline{x}_r$ which form part of a regular system of parameters of $\overline{S}$, see Algebra, Lemma 00NR. Lift these elements to $x_1, \ldots, x_r \in S$. Then $t_1, \ldots, t_d, x_1, \ldots, x_r$ is part of a regular system of parameters for S. Hence $S/(x_1, \ldots, x_r)$ is a power series ring over a field (if the characteristic of k is zero) or a power series ring over a Cohen ring (if the characteristic of k is $p > 0$), see Lemma 07NM. Moreover, it is still the case that $R \to S/(x_1, \ldots, x_r)$ maps $t_1, \ldots, t_d$ to a part of a regular system of parameters of $S/(x_1, \ldots, x_r)$. In other words, we may replace S by $S/(x_1, \ldots, x_r)$ and assume we have a diagram

$$\begin{array}{ccc} S & \longrightarrow & B \\ \uparrow & & \uparrow \\ R & \longrightarrow & A \end{array}$$

as in Lemma 07NN with moreover $\overline{S} = B$. In this case $R \to S$ is formally smooth in the $\mathfrak{m}_S$-adic topology by Proposition 07NQ. Hence the base change $C = S \otimes_R A$ is formally smooth over A in the $\mathfrak{m}_C$-adic topology by Lemma 07EG. □